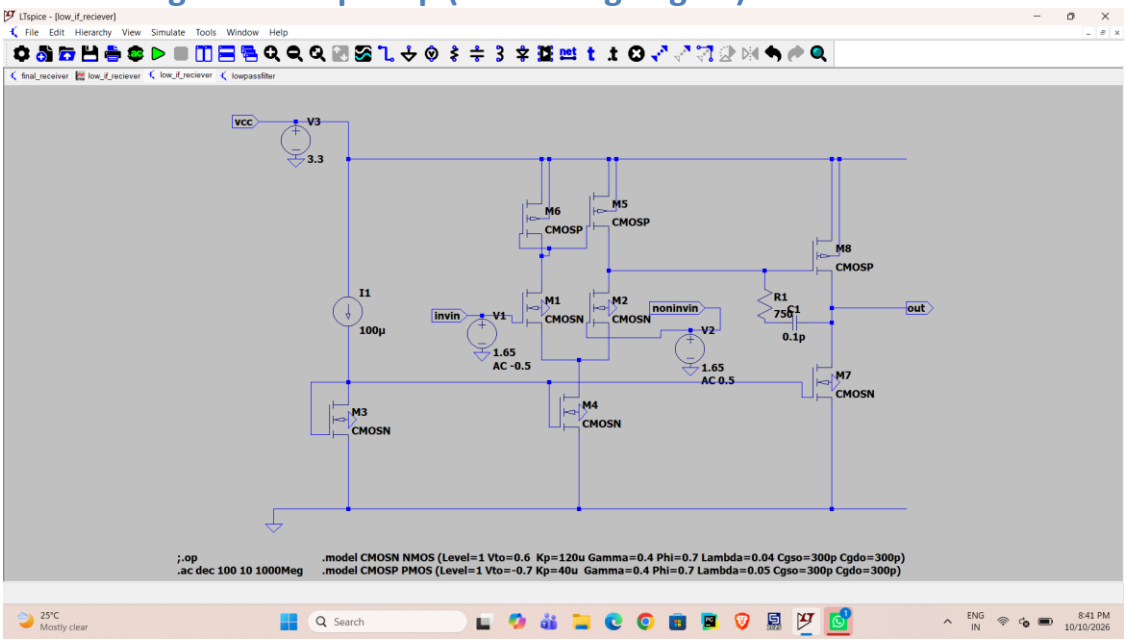
Circuit Diagrams, Design Values and Component Roles

A. Two-stage CMOS op-amp (the analog engine)



Op-amp schematic

The schematic sheet is turned into a hierarchical block with ports `invin`, `noninvin`, `out`, `vcc`. The sources V1/V2 (AC ±0.5, 1.65 V DC) are test sources used for the AC analysis in the stand-alone sheet.

Device list and why these values

Process model (generic Level-1, L = 1 µm for every device): NMOS K_p = 120 µA/V², V_{to} = 0.6 V, λ = 0.04; PMOS K_p = 40 µA/V², V_{to} = −0.7 V, λ = 0.05.

Device	Type, W/L (µm)	Role	Why this value
I1	100 µA source	Bias reference	Sets all currents through mirrors
M3	NMOS 20/1	Bias diode	Converts I1 into the bias voltage for M4 and M7
M4	NMOS 20/1	Tail current source	Mirrors I1: 100 µA shared by the input pair
M1, M2	NMOS 20/1	Input differential pair	50 µA each. From $I_D = ½ K_p (W/L) V_{ov}^2$: $W/L \approx 21$ gives $V_{ov} \approx 0.2$ V and $g_m \approx 0.5$ mS
M5, M6	PMOS 28/1	Current-mirror load	50 µA each at $V_{ov} \approx 0.3$ V ($W/L = 2I_D/(K_p V_{ov}^2) \approx 28$)
M8	PMOS 112/1	Second-stage common-source amplifier	4 × M5/M6 for 200 µA, so $V_{SG8} = V_{SG6}$ and the output is balanced (no systematic offset)
M7	NMOS 40/1	Second-stage current sink	2 × M4 for 200 µA; acts as the active load of M8
C1 (Cc)	0.1 pF	Miller compensation	Creates the dominant pole, sets gain-bandwidth ≈ $g_{m1}/(2\pi C_c)$
R1 (Rz)	750 Ω	Nulling resistor	≈ $1/g_{m8}$ ($g_{m8} \approx 1.33$ mS); cancels the right-half-plane zero from Cc
V3	3.3 V	Supply	Single supply

Expected performance (ngspice check)

Item	Value
Stage 1 gain (diff pair)	about 118 V/V (41.5 dB)
Stage 2 gain	about 74 V/V (37 dB)
Total DC gain	about 9700 V/V (79.7 dB)
Supply current	405 µA (about 1.3 mW)

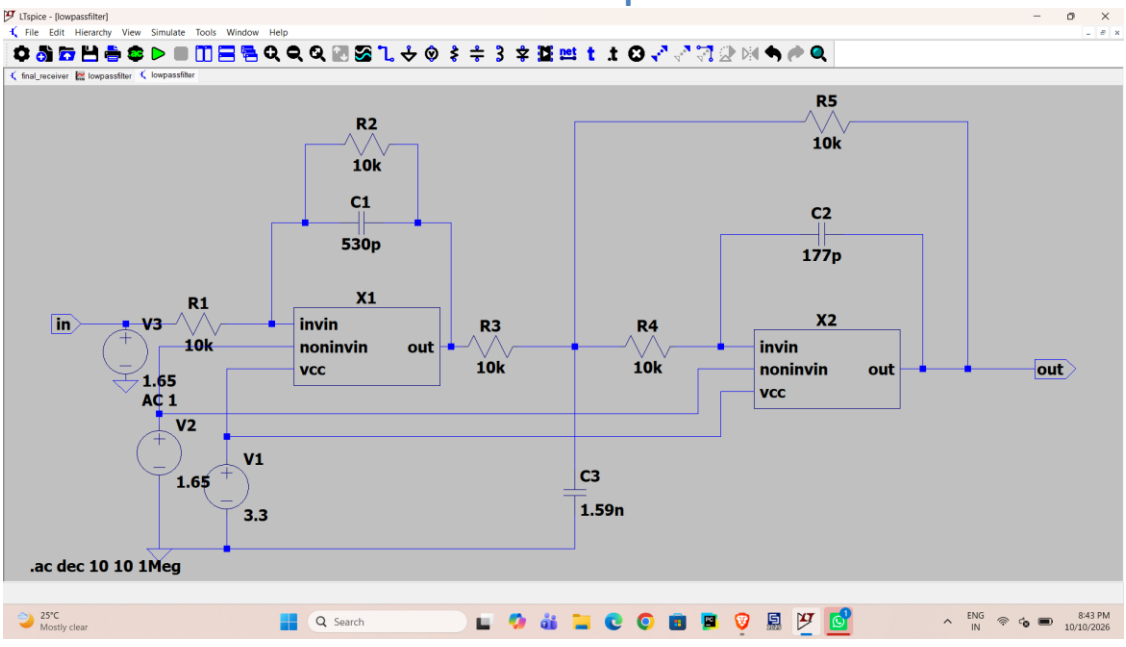
Compensation trade-off

Unity-gain phase margin versus Miller capacitor (*ngspice*):

Cc	Unity-gain frequency	Phase margin
0.1 pF (used)	206 MHz	about 13°
0.3 pF	135 MHz	about 30°
0.5 pF	103 MHz	about 40°
1 pF	65 MHz	about 56°
2 pF	37 MHz	about 70°

Cc = 0.1 pF maximizes gain-bandwidth so the active-RC sections track their ideal response (earlier experiments with a 1 MHz IF showed that a low-GBW op-amp raises the filter Q). The unity-gain margin is low, but the filters use feedback factors well below 1 and the results show no ringing. For this 30 kHz filter, 0.5 to 1 pF would give more margin.

B. 3rd-order Butterworth active-RC low-pass filter



Low-pass filter schematic

The filter is two op-amp stages: a first-order inverting section (X1) followed by a multiple-feedback (MFB) biquad (X2). Both inverting stages cancel, so the DC gain is +1. The non-inverting inputs are tied to V_{ref} = 1.65 V (V2).

Transfer function

A 3rd-order Butterworth filter has one real pole and a complex pair with $Q = 1$:

$$H(s) = \frac{1}{1 + s/\omega_c} \cdot \frac{\omega_c^2}{s^2 + \omega_c s + \omega_c^2}, \qquad \omega_c = 2\pi \cdot 30\ \text{kHz}$$

Component values

Part	Value	Role
R1	10 kΩ	Input resistor of section 1 (sets gain -R2/R1 = -1)
R2	10 kΩ	Feedback resistor of section 1
C1	530 pF	Across R2: pole at $1/(2\pi R_2 C_1)$ = 30.0 kHz
R3	10 kΩ	Input resistor of the biquad
R5	10 kΩ	Feedback resistor from the biquad node to the output
R4	10 kΩ	Resistor from the biquad node to the inverting input
C3	1.59 nF	Biquad node to ground: sets the pole frequency
C2	177 pF	Op-amp feedback capacitor: sets the damping (Q)
V2	1.65 V	Reference for the non-inverting inputs (mid-supply)
V1	3.3 V	Supply

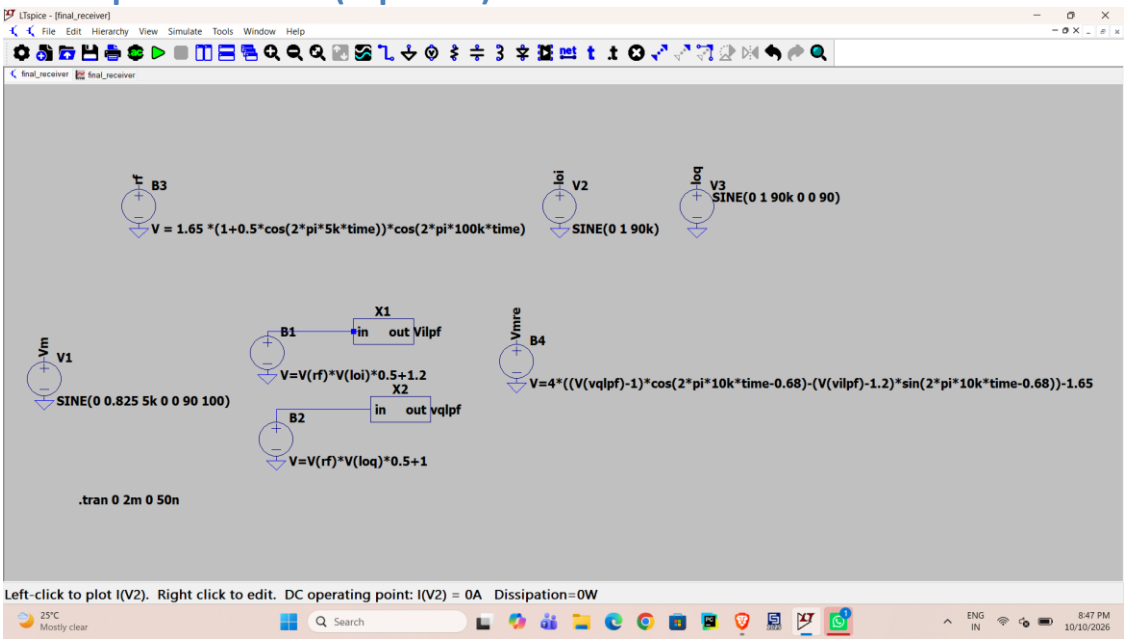
Why these values

For the MFB section with all resistors equal to R and unity gain:

$$\omega_0 = \frac{1}{R\sqrt{C_2 C_3}}, \qquad Q = \frac{1}{3}\sqrt{\frac{C_3}{C_2}}$$

- $Q = 1$ requires $C_3 = 9\ C_2$.
- With R = 10 kΩ and $\omega_0 = 2\pi \cdot 30$ kHz: $C_2 = 1/(3R\ \omega_0)$ = 177 pF and $C_3 = 9\ C_2$ = 1.59 nF.
- First section: $C_1 = 1/(2\pi R \cdot 30\ \text{kHz})$ = 530 pF.
- **Why 30 kHz:** the IF is 10 kHz and the message sidebands extend to 15 kHz. A 30 kHz cutoff passes both with about 0.1 dB loss (gain 0.988 at 15 kHz) and rejects the 190 kHz mixer sum product by 48 dB. A 100 kHz cutoff left too much of the sum term, and mismatched capacitors (a cutoff near 11 kHz) attenuated the IF itself.
- **Why 10 kΩ:** low enough for modest thermal noise and small capacitors, high enough that the op-amp can drive it comfortably.
- **Why a Butterworth response:** maximally flat passband, so the IF and its sidebands are not distorted.

C. Complete receiver (top level)



Receiver schematic

Block	Expression / value	Purpose
V1 (vm)	`SINE(0 0.825 5k 0 0 90 100)`	Reference message, 0.825 cos(2π·5k·t)
B3 (rf)	`1.65*(1+0.5*cos(2*pi*5k*time))*cos(2*pi*100k*time)`	AM RF signal: carrier 1.65 V, μ = 0.5
V2 (loi)	`SINE(0 1 90k)`	In-phase LO: sin(2π·90k·t)
V3 (loq)	`SINE(0 1 90k 0 0 90)`	Quadrature LO: cos(2π·90k·t)
B1	`V(rf)*V(loi)*0.5+1.2`	I mixer: product × 0.5, plus 1.2 V offset
B2	`V(rf)*V(loq)*0.5+1`	Q mixer: product × 0.5, plus 1.0 V offset
X1, X2	`lowpassfilter` (section B)	3rd-order Butterworth, 30 kHz, one per path
B4 (vmre)	`4*((V(vqlpf)-1)*cos(2*pi*10k*time-0.68)-(V(vilpf)-1.2)*sin(2*pi*10k*time-0.68))-1.65`	Second quadrature mixer, scaling and DC removal
B5	`sqrt(V(Vm)^2)`	Unused test source
`.tran`	`0 2m 0 50n`	2 ms simulation, 50 ns maximum step

Why these values

- **LO 90 kHz, IF 10 kHz.** An IF different from the message frequency avoids confusing the IF beat with the message (with an IF equal to 5 kHz the “recovered” tone was the IF itself, not the message). 10 kHz also keeps the 5 kHz sidebands (5 to 15 kHz) clear of DC.
- **Mixer scale 0.5 and offsets 1.2 V / 1.0 V.** The scale keeps the raw product (up to ±1.24 V) inside the op-amp range. The offsets center each path in the 0 to 3.3 V window. The first filter stage output sits at 3.3 − V_{offset} (2.1 V for I, 2.3 V for Q) and swings about ±0.6 V, so the Q path is the

closer of the two to the upper limit of the op-amp. Using 1.2 V for both paths would give the Q path more headroom (then replace `-1` by `-1.2` in B4).

- **Phase 0.68 rad.** Equals the filter phase lag at 10 kHz (39°). Without it the output is scaled by cos 39° = 0.78.
- **Gain 4 and -1.65 V.** The factor 4 cancels the mixer gain (0.5) and the 1/2 from the product-to-sum identity, giving $A_c + m(t)$; subtracting A_c = 1.65 V leaves $m(t)$.
- **Reference message.** `V(vm)` is plotted only to compare with `V(vmre)`.

D. Component significance, summary

Component	Significance
Differential pair (M1, M2)	Amplifies the voltage difference; sets g_m and so the gain-bandwidth
Current mirror (M5, M6)	Converts the differential current to a single-ended output and doubles the gain
Tail source (M4)	Sets total input-stage current and rejects common-mode signals
Common-source stage (M8, M7)	Provides the second gain stage and a larger output swing
Miller capacitor (Cc)	Dominant pole: makes the amplifier stable in feedback
Nulling resistor (Rz)	Removes the right-half-plane zero that would reduce phase margin
Filter capacitors C1, C2, C3	Set the pole frequencies and the damping of the Butterworth response
Filter resistors (10 kΩ)	Set the gains (unity) and, with the capacitors, the pole frequencies
Reference 1.65 V	Biases the op-amps at mid-supply so signals can swing both ways on one supply
I/Q mixers	Translate the RF band down to the IF in two quadrature paths
Second quadrature mixer	Combines I and Q to remove the IF carrier and reconstruct the message